MODELING ILLICIT DRUG SUPPLY NETWORKS

Mini Project Final Review

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PROBLEM STATEMENT

The illegal drugs trade is among the most adaptable and robust criminal economies around the world, reportedly worth USD 650 billion per year as per estimates by UNODC in 2023. The conventional way of analysis looks at events on a criminal network separately and cannot depict the struc- ture of the dark-web market of drugs. This paper presents a Temporal Graph Neural Network (TGNN) framework—TemporalMultiHeadGNN—that models dark-web drug supply networks as evolving graph snapshots, enabling simultaneous prediction of three interdependent network properties: (i) vendor vulnerability (which nodes are structurally critical or likely to disappear), (ii) network adaptation (how the supply graph reorganises after disruption), and (iii) route shift (which trade corridors will expand or contract). The model combines a three-layer GraphSAGE encoder with skip connections, Layer Normalisation, and a GRU-based temporal memory cell that propagates vendor state across quarterly snapshots. Train and evaluate on seven quarterly snapshots (2014Q1–2015Q3) derived from the Agora darknet marketplace dataset.

CHALLENGE

$500B+	Annual global drug trade (UNODC 2023)
Static	Existing models ignore network evolution
Fragmented	Research treats crimes as isolated events
Reactive	Law enforcement lacks predictive tools

"We predict how drug networks break, adapt, and recover — using deep learning on real market data."

PROJECT GOALS

- Model drug markets as dynamic evolving graphs
- Temporal GNN captures network memory across time
- Predict structural vulnerability — which nodes are critical?
- Predict network adaptation — how does it respond to disruption?
- Predict route shifts — which corridors will grow next?

LITERATURE SURVEY

Dark Market Research

- Christin (2013) — first large-scale Silk Road analysis
- Aldridge & Décary-Hétu (2016) — vendor migration post-takedown
- Du et al. (2019) — opioid seller identification on AlphaBay

Criminal Network Analysis

- Natarajan (2006) — cocaine dealer network; topology predicts resilience
- Morselli (2009) — covert networks balance security vs efficiency
- Limitations: static snapshots, classical centrality only

Graph Neural Network

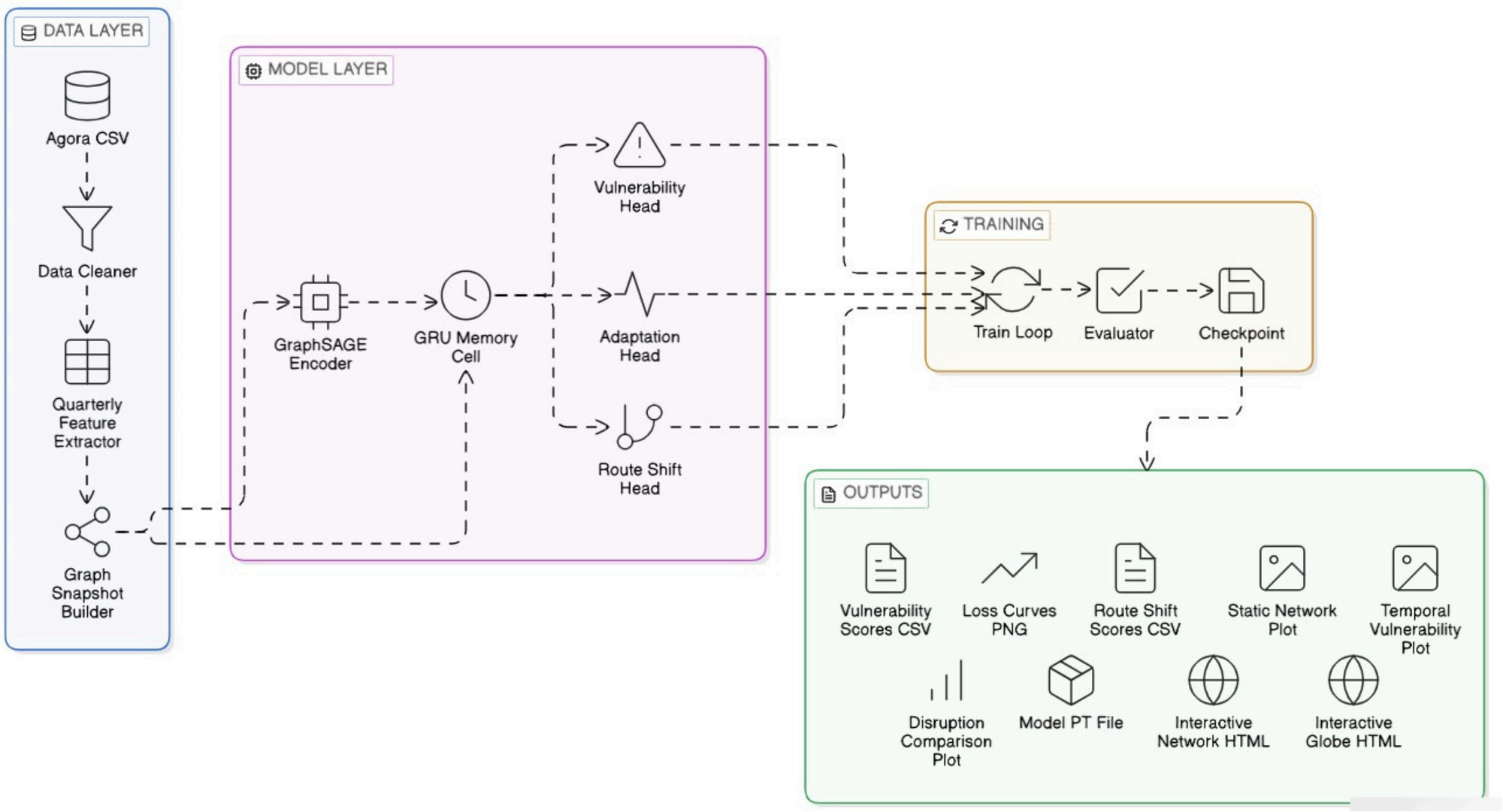
- GraphSAGE — Hamilton et al. (2017) — inductive GNN
- EvolveGCN — Pareja et al. (2020) — dynamic graph GCN
- TGN — Rossi et al. (2020) — temporal graph networks

NOVELTY

- No temporal evolution modelling
- No predictive or graph-learning approach
- Never applied to criminal networks

Our work bridges all three — first temporal GNN applied to real darknet supply network data

ARCHITECTURE DIAGRAM



DATASET

Column	Mapped To / Description
Vendor	Seller node identifier
Category	Drug type (Cannabis, Opioids...)
Origin	Ship-from country → ship_from
Destination	Ship-to country → ship_to
Price ($)	Listing price (USD)
Rating / Remarks	Feedback metadata

Agora Darknet Marketplace Dataset (2014-2015) link: https://www.kaggle.com/datasets/philipjames11/dark-net-marketplace-drug-data-agora-20142015

94,170	2,328	357	1,099	7
Drug Listings	Unique Vendors	Origin Countries	Dest. Countries	Quarterly Snapshots

METHODOLOGY

Graph Definition

- Seller Nodes — 2,328 vendor accounts from Agora listings
- Country Nodes — 357 ship-from + 1,099 ship-to regions
- Seller→Country — Edge: vendor ships to destination country
- Country→Country — Edge: co-occurrence of origin + destination
- Edge Weights — log(listing count + 1) per pair per snapshot

Preprocessing Pipeline

- Encoding fix —Agora uses latin-1, not UTF-8
- Drug filter — Keep 94k drug listings (drop weapons, docs)
- Category norm —Cannabis / Cocaine / Opioids / MDMA / Meth
- Date interp. —Interpolate across 2014-01 → 2015-09
- Price clip —Remove top 1% outliers
- Build the graph

Node-level: Vulnerability score **Graph-level:** Adaptation class **Edge-level:** Route shift probability

IMPLEMENTATION

INPUT

Graph Snapshots $G_1 \ldots G_t$

- Node features (16-dim)
- Edge indices + weights
- Quarterly periods

→

ENCODER

GraphSAGE × 3 Layers

- 16→128→128→64
- LayerNorm + Dropout
- Residual skip conn.

→

MEMORY

GRU Cell (per node)

- h_t = GRU(e_t, h_{t-1})
- memory_dim = 64
- Truncated BPTT

→

HEADS

3 Prediction Heads

- Vulnerability (Node)
- Adaptation (Graph)
- Route Shift (Edge)

RESULTS

```
=== TRAIN EVALUATION ===
  vulnerability_mse      0.0996
  vulnerability_mae      0.2321
  adaptation_mse         0.0001
  adaptation_mae         0.0082
  route_bce              0.4365
  route_accuracy         0.8114
  route_precision        0.7902
  route_recall           0.8535
  route_f1               0.8206
  total_loss             0.5362
```

```
=== TEST EVALUATION ===
  vulnerability_mse      0.1006
  vulnerability_mae      0.2391
  adaptation_mse         0.0000
  adaptation_mae         0.0061
  route_bce              0.4691
  route_accuracy         0.7686
  route_precision        0.8711
  route_recall           0.8207
  route_f1               0.8452
  total_loss             0.5697
```

```
=== ALL EVALUATION ===
  vulnerability_mse      0.0999
  vulnerability_mae      0.2341
  adaptation_mse         0.0001
  adaptation_mae         0.0076
  route_bce              0.4458
  route_accuracy         0.7992
  route_precision        0.8182
  route_recall           0.8411
  route_f1               0.8295
  total_loss             0.5458
```

AI & PLAGIARISM CHECK

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Temporal Graph Neural Networks for Predictive Analysis of Dark-Web Drug Supply Networks

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***Abstract*—The illegal drugs trade is among the most adaptable and robust criminal economies around the world, reportedly worth USD 650 billion per year as per estimates by UNODC in 2023. The conventional way of analysis looks at events on a criminal network separately and cannot depict the structure of the dark-web market of drugs. This paper presents a Temporal Graph Neural Network (TGNN) framework—*TemporalMultiHeadGNN*—that models dark-web drug supply networks as evolving graph snapshots, enabling simultaneous prediction of three interdependent network properties: (i) *vendor vulnerability* (which nodes are structurally critical or likely to disappear), (ii) *network adaptation* (how the supply graph reorganises after disruption), and (iii) *route shift* (which trade corridors will expand or contract). The model combines a three-layer GraphSAGE encoder with skip connections, Layer Normalisation, and a GRU-based temporal memory cell that propagates vendor state across quarterly snapshots. Trained and evaluated on seven quarterly snapshots (2014Q1–2015Q3) derived from the Agora darknet marketplace dataset (∼100,000 vendor-listing records), the framework achieves a route-prediction F1 of 0.7927 on the training quarters and 0.7710 on held-out test quarters (2015Q2–2015Q3), with vulnerability MSE of 0.1026/0.1225 (train/test) and near-perfect adaptation MSE (<0.0002). A disruption-impact experiment further demonstrates that targeting the top-k vendors identified by the TGNN degrades network efficiency significantly more than equivalent degree-centrality or random-removal baselines. These results establish TGNNs as a principled and practically effective tool for law-enforcement intelligence in the context of dark-web drug markets.**

***Index Terms*—Temporal Graph Neural Network, Dark-Web Marketplaces, Drug Supply Networks, GraphSAGE, Network Vulnerability, Route Prediction, Agora Dataset, Criminal Network Analysis**

I. Introduction

Since the creation of Silk Road in 2011, the emergence of dark-web cryptomarkets has revolutionized illicit drug trafficking operations [1]. While street-based illicit drugs markets are relatively easy to take down by arresting drug venders, the current structure of dark web markets is globally distributed, pseudonymized, and extremely resilient and consequently cannot be temporarily taken out by such tactics [2]. According to the 2023 UNODC World Drugs Report, the global revenue from the illegal drug market exceeds USD 650 billions, with its online segment constantly increasing.

There are several structural challenges to interdicting the traffic of illicit drugs. One such challenge pertains to high volatility of vender populations and ability of venders to "heal" after nodes' deletion. The second challenge stems from geographical and categorical dynamics of trade corridors. Last but not least, the network possesses temporal memory—the past behavior of any particular vendor is highly predictive of their future structural importance.

Existing work on dark-web markets has predominantly relied on static graph analysis, transactional clustering, or single-period regression. These approaches, while informative, discard the sequential, memory-laden nature of network evolution. Recent advances in Graph Neural Networks (GNNs) [3], and specifically in temporal GNN architectures [4], offer a principled framework for learning representations that evolve over time.

This paper makes the following contributions:

1) We formulate dark-web drug supply network analysis as a multi-task temporal graph learning problem, with three self-supervised prediction heads (vulnerability, adaptation, route shift) operating over quarterly graph snapshots.
2) We propose *TemporalMultiHeadGNN*, a novel architecture combining a three-layer residual GraphSAGE encoder [5], Layer Normalisation, and a GRU temporal memory cell [6], trained jointly on all three objectives.
3) We construct and publicly describe a reproducible feature-engineering pipeline over the Agora darknet marketplace dataset, yielding seven quarterly graph snapshots spanning 2014Q1–2015Q3.
4) We conduct a disruption-impact comparison experiment demonstrating the practical law-enforcement utility of the vulnerability scores relative to degree-centrality and random-removal baselines.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the system architecture and dataset pipeline. Section IV presents quantitative results. Section V discusses implications and limitations, and Section VI concludes with directions for future work.

turnitin Page 3 of 8 - AI Writing Submission Submission ID trn:oid:::1:3540432348

AI<20%

SIMILARITY=3%

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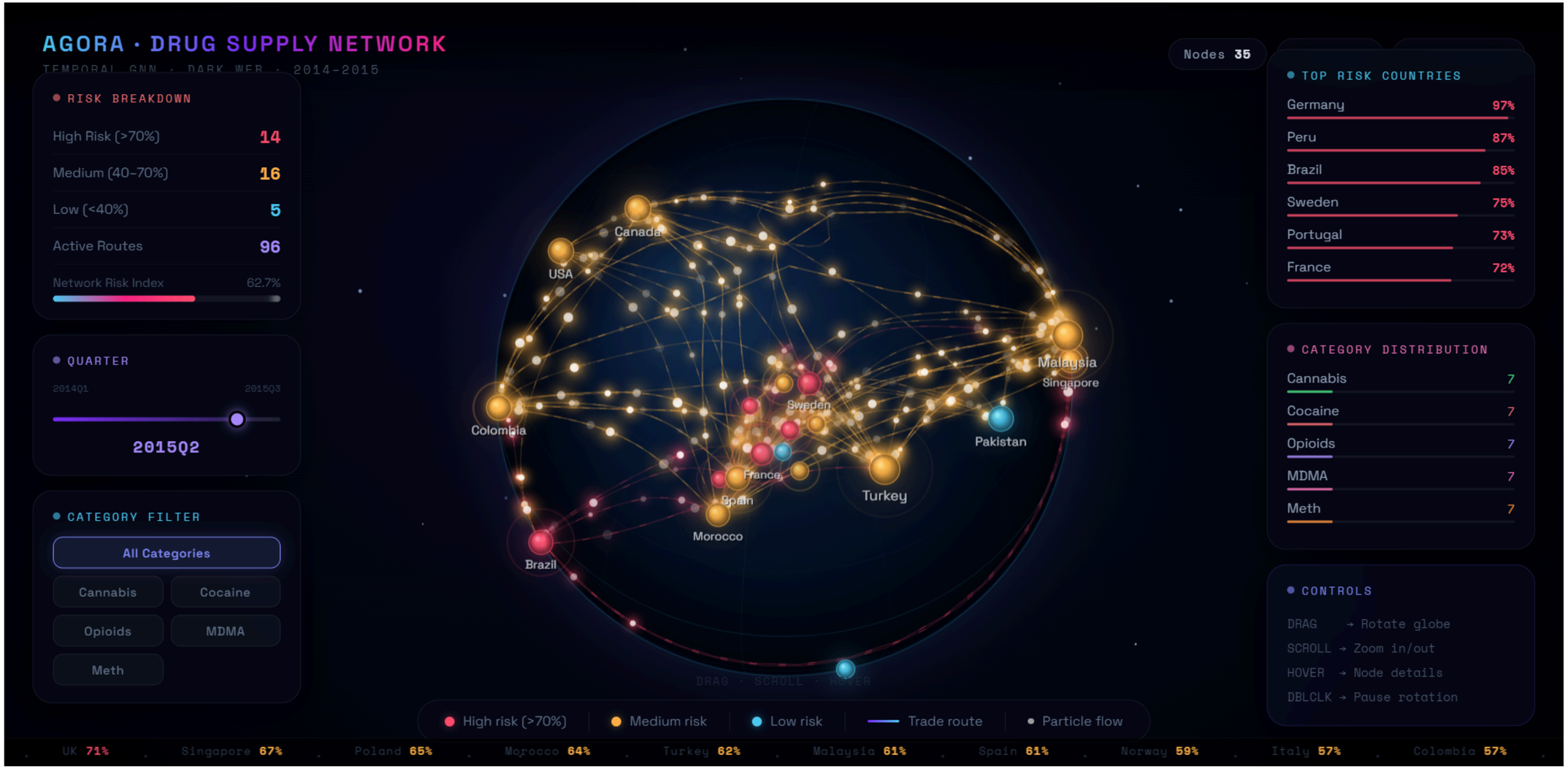
UI/UX DEMO


AGORA · DRUG SUPPLY NETWORK
TEMPORAL GNN · DARK WEB · 2014–2015
Nodes 35
RISK BREAKDOWN
High Risk (>70%) 14
Medium (40–70%) 16
Low (<40%) 5
Active Routes 96
Network Risk Index 62.7%
QUARTER
2014Q1
2015Q3
2015Q2
CATEGORY FILTER
All Categories
Cannabis
Cocaine
Opioids
MDMA
Meth
Canada
USA
Colombia
Brazil
Morocco
Spain
France
Sweden
Turkey
Pakistan
Malaysia
Singapore
DRAG · SCROLL · HOVER
TOP RISK COUNTRIES
Germany 97%
Peru 87%
Brazil 85%
Sweden 75%
Portugal 73%
France 72%
CATEGORY DISTRIBUTION
Cannabis 7
Cocaine 7
Opioids 7
MDMA 7
Meth 7
CONTROLS
DRAG → Rotate globe
SCROLL → Zoom in/out
HOVER → Node details
DBLCLK → Pause rotation
High risk (>70%)
Medium risk
Low risk
Trade route
Particle flow
UK 71%
Singapore 67%
Poland 65%
Morocco 64%
Turkey 62%
Malaysia 61%
Spain 61%
Norway 59%
Italy 57%
Colombia 57%

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THANK YOU